LECTURER: JOHN DOE

REINFORCEMENT LEARNING

Who are you?

- Name
- Employer
- Position/responsibilities
- Fun Fact
- Previous knowledge? Expectations?



REINFORCEMENT LEARNING
TOPIC OUTLINE

Introduction to Reinforcement Learning	1
Dynamic Programming and Markov Decision Processes	2
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UNIT 1

INTRODUCTION TO REINFORCEMENT LEARNING

STUDY GOALS



- Understand **what is reinforcement learning** and identify the various problems addressed by it

- Compare and contrast **how reinforcement learning surpasses challenges** encountered by traditional control theory methods

- Categorize the different **approaches** employed **in reinforcement learning**

EXPLAIN SIMPLY



1. Explain the relationship between reinforcement learning and dynamic programming to address control problems

2. Describe the key components of reinforcement learning systems, including policy evaluation and iterative updating

3. Compare on-policy and off-policy reinforcement learning methods

WHY REINFORCEMENT LEARNING



Source of the text: James, Witten, Hastie, & Tibshirani, 2013.

Machine Learning: teaching computers to learn and decide from data autonomously

Supervised learning

- Learn with a teacher
- Train using input-ouput pairs
- Goal: classification



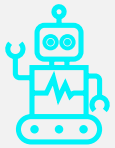
Unsupervised learning

- Find patterns in data
- No labeled data, only features
- Goal: grouping



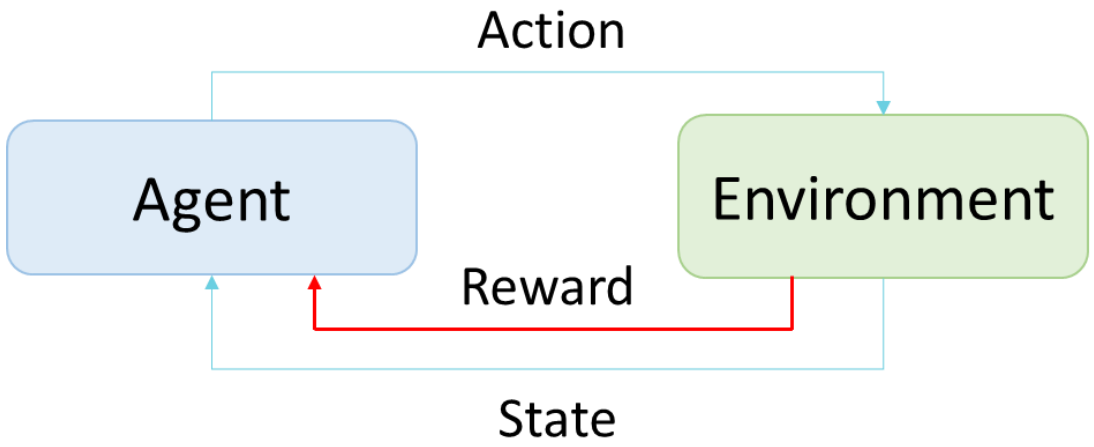
Reinforcement learning

- Trial and error
- Agents act in environment
- Goal: learn to act



Optimizing computer decisions through trial-and-error learning

- **State:** current condition
- **Action:** allowed moves in each state
- **Reward:** incentive for actions
- **Policy:** map states to actions



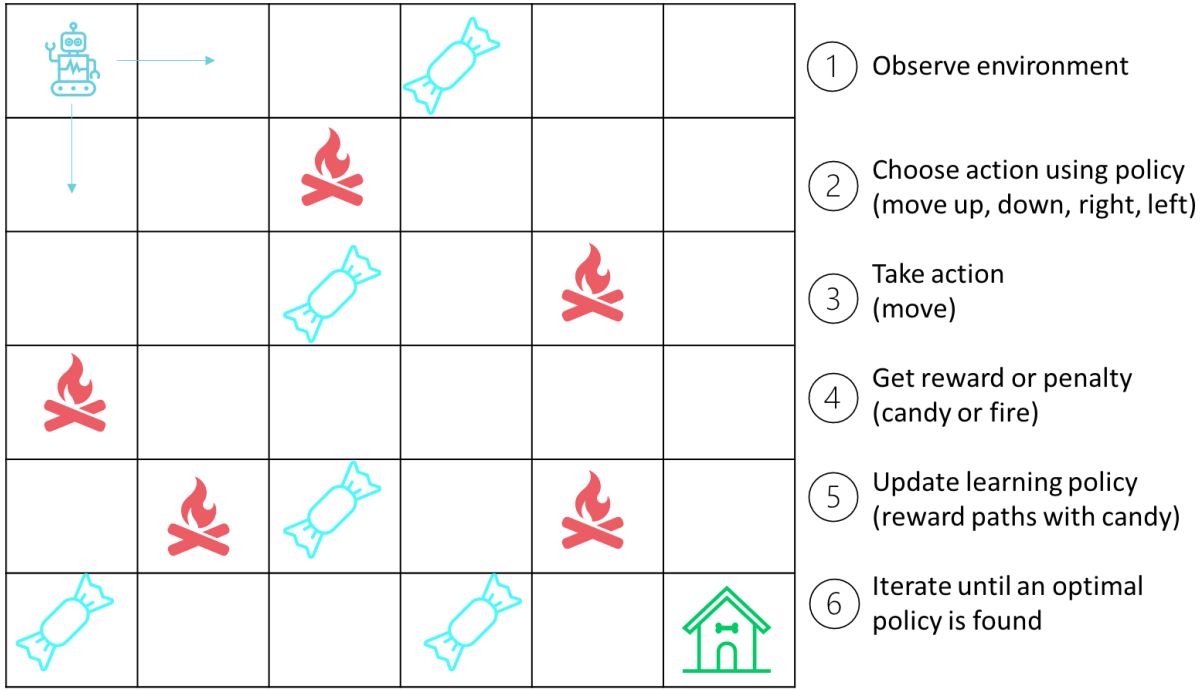
- **Agent**: the learning entity
- **Environment**: external context

Source of the text: Sutton & Barto, 2018.
Source of the image: Own representation.

Markov Decision Process (MDP)

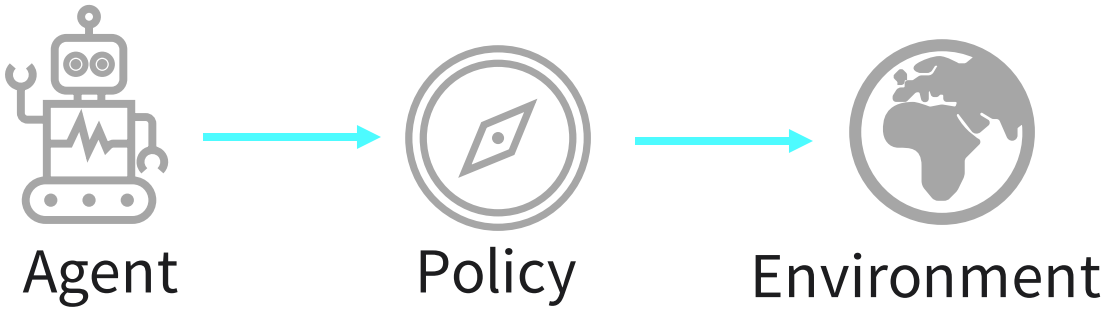
Framework to model sequential decision-making under uncertainty

- State-action space K
- Cost function $\mathrm{R}(\mathrm{x}, \mathrm{a})$
- Probability distribution $\mathrm{P}(\mathrm{y} \mid \mathrm{x}, \mathrm{a})$
- Discounted cost / reward function β



Source of the image: Own representation.

COMPONENTS OF REINFORCEMENT LEARNING



$$V^*(x) = \min_{a \in A_x} R(x,a) + \beta \sum_{y \in X} V^*(y) P(y \mid x,a)$$

- Policy maps states to actions
- Agent learns policies by experience
- Optimal policy = highest reward
- **Bellman equation:** guides policy improvement and optimization
- Evaluates state-action pairs
- Cornerstone of reinforcement learning

Source of the text: Sutton & Barto, 2018.

FUNDAMENTAL REINFORCEMENT LEARNING CATEGORIES AND PRINCIPLES

Approaches

Model free

- Learn from interactions; no explicit model
- Q-learning for action-value estimation

Model based

- Use explicit model for planning
- Monte Carlo Tree Search (MCTS)

VS

On policy

- Improve using own policy's data
- SARSA for online learning

Off policy

- Learn from different policy's data
- Q-learning for off-policy learning

Categories

Source of the text: Sutton & Barto, 2018.

POLICY EVALUATION AND POLICY IMPROVEMENT



Refine policy through value assessments and action enhancements

Policy evaluation

- Initialize value of state V(s)
- Update V(s) using Bellman equation
- Repeat until convergence

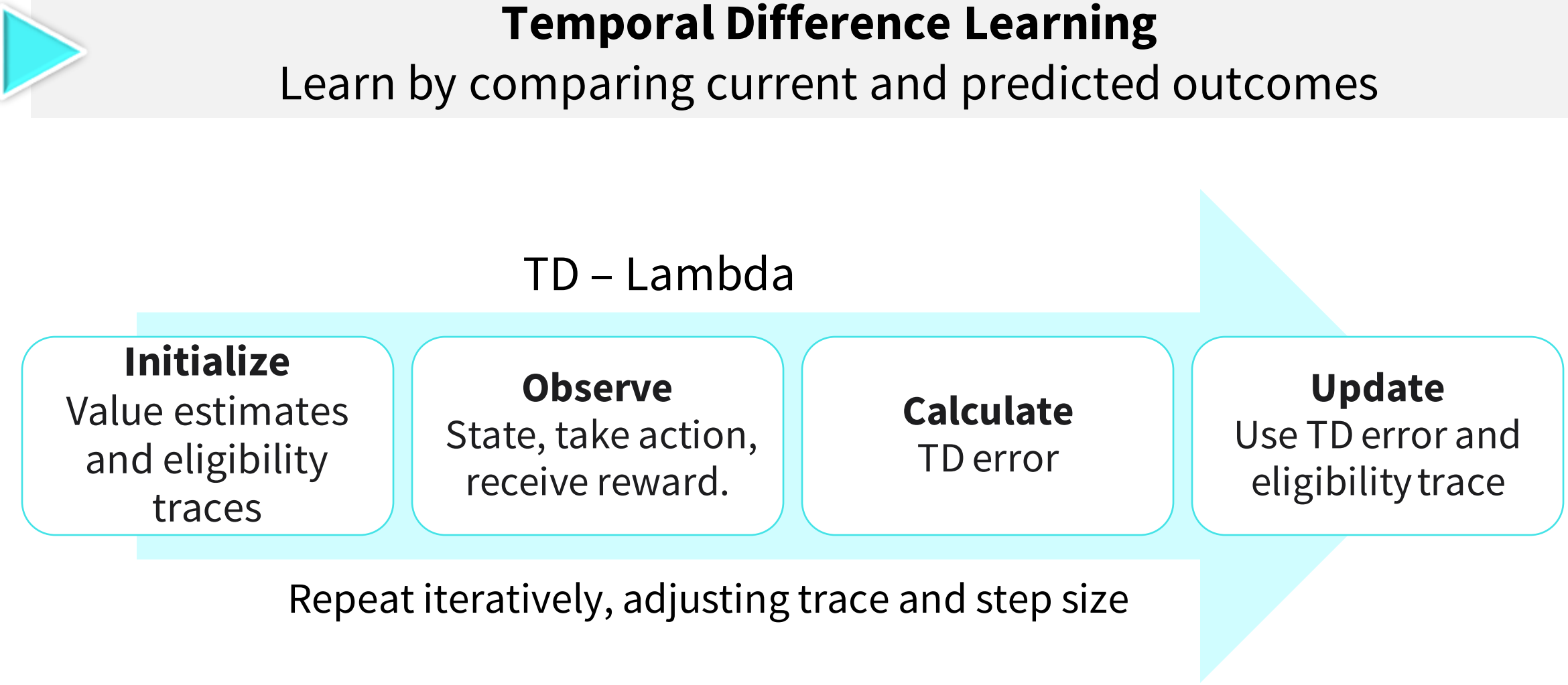
Policy improvement

- Get best estimate for each state
- Look ahead to find best policy
- Pick action with highest reward

Source of the text: Sutton & Barto, 2018.

Temporal Difference Learning

Learn by comparing current and predicted outcomes



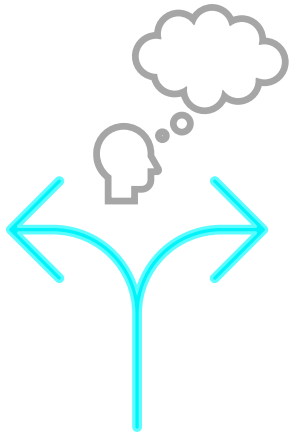
Optimize an agent's actions by learning from experiences

Q- Learning (off-policy)

- **Explore**: Collect experience using exploration
- **Update**: Update Q-values using Bellman equation
- **Exploit**: Choose actions based on updated Q-values.
- **Iterate**: Repeatedly learn and adjust Q-values.

SARSA (on-policy)

- **Explore**: Epsilon-greedy action selection
- **Act**: Perform chosen action
- **Receive**: Get reward, observe new state
- **Update**: Adjust Q-value based on observed outcome.



Exploitation

- Choose high reward actions
- Exploit learned information
- Maximize immediate gain

Exploration

- Discover novel actions
- Learn about environment
- Balance knowledge expansion

When to explore new options and when to exploit what we know?

REVIEW STUDY GOALS



- Understand **what is reinforcement learning** and identify the various problems addressed by it

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SESSION 1

TRANSFER TASK

Case study

An e-commerce giant wants to optimize its warehouse operations using autonomous robots. The robots need to navigate through the warehouse, locate items, avoid obstacles, and efficiently transport packages to packing stations

Task

Explain how this situation can be translated into a reinforcement learning problem. Define the key components required for modeling this task

TRANSFER TASK
PRESENTATION OF THE RESULTS

Please present your results.

The results will be discussed in plenary.





1. A reinforcement learning application is designed to play a card game automatically. A play holds five cards from a standard deck of 52. At each stage the player can discard and replace zero, one or two cards. How many actions are available at each stage
   a) 2
   b) 10
   c) 16
   d) 26



2. Which statement best describes the exploration-exploitation dilemma?
   a) This occurs when the decision maker wishes to both apply the best available policy and collect new data in order to improve the policy
   b) This arises because determination of the optimal policy requires statistical estimation
   c) If the environment changes, the current optimal policy is no longer optimal
   d) In general, a reinforcement learning application must visit every state-action pair infinitely often in order to converge to the optimal policy



3. If for a minimum cost MDP the discount rate is $\beta = 0.0$, how would an optimal policy be calculated
   a) The total discounted cost would always be zero, so any policy would be optimal
   b) From any state x the optimal action a would be the one minimizing R(x,a), where x is held fixed
   c) By using value iteration after setting R(x,a) to zero
   d) The problem is not well-defined, so no optimal policy exists

LIST OF SOURCES

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An introduction to statistical learning* (Vol. 112, p. 18). New York: Springer.

Sutton, R. S., & Barto, A. G. (2018a). Reinforcement learning: An introduction (2nd ed.). MIT press.

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